

## Section - A.

MCQs :-

- 1) C
- 2) B
- 3) B
- 4) B
- 5) C
- 6) A
- 7) B
- 8) ~~A~~ C
- 9) C
- 10) D
- 11) B
- 12) C



## Section - C

Q:3 (i) (or)

A:3 (i) (or)

(i) ~~Rules~~ Assigning rules for oxidation numbers or states :-

- 1) The oxidation state of any uncombined or free is always zero. e.g. oxidation n.o of  $S_8$ ,  $H_2$ ,  $N_2$ ,  $I_2$ , etc is always zero.
- 2) In simpler ions, the oxidation number is same as their charges. e.g. oxidation number of Na in  $Na^+$  and oxidation number of Ca in  $Ca^{+2}$  respectively is +1 and +2.
- 3) In complex ion, the sum of the oxidation numbers of the atoms is equal to their charges on their ions. e.g. in  $CO_3^{2-}$ , the sum of oxidation states of one Carbon and 3 oxygen atoms is -2. Similarly  $NH_4^+$ , the sum of oxidation states of one nitrogen and 4 hydrogen atoms is always +1.



i) The oxidation numbers of each of the atoms in a molecule or compound is counted ~~seer~~ separately or their algebraic sum is always zero. e.g. HCl, the sum of oxidation state of one hydrogen and one ~~ee~~ chlorine atoms is 0. Similarly in  $\text{CO}_2$ , the sum of oxidation state of one C and 2 Oxygen atoms is always zero.

ii) Oxidation state of :-

1) B in  $\text{H}_3\text{BO}_3$

$$1(3) + x + 3(-2) = 0$$

$$\Rightarrow 3 + x - 6 = 0$$

$$x - 6 + 3 = 0$$

$$x - 3 = 0$$

$$x = +3$$

3) Mn in  $\text{KMnO}_4$

$$1 + x + 4(-2) = 0$$

$$1 + x - 8 = 0$$

$$x - 8 + 1 = 0$$

$$x - 7 = 0$$

$$x = 0 + 7$$

$$x = +7$$

2) N in  $\text{NH}_4^{+1}$

$$x + 4(1) = +1$$

$$x + 4 = 1$$

$$x = 1 - 4$$

$$x = -3$$



Q.3 (ii)

A.3 (ii)

### \* Oxidizing agents :-

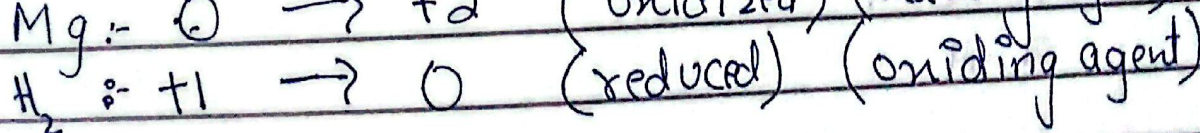
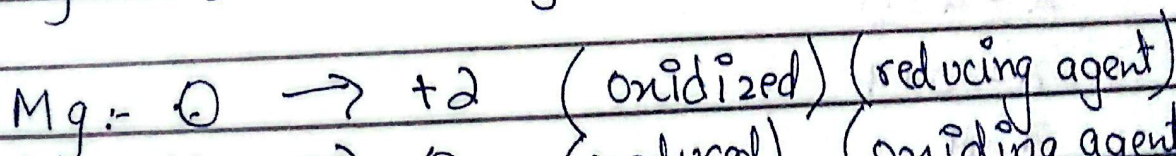
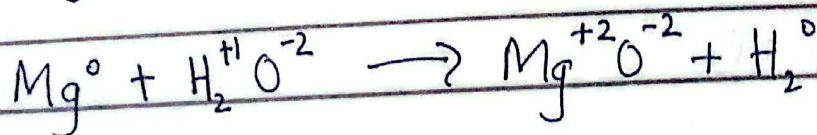
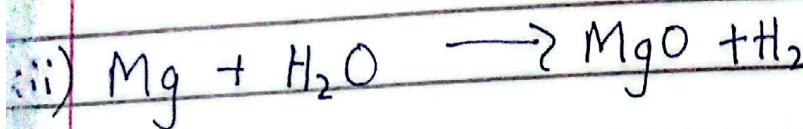
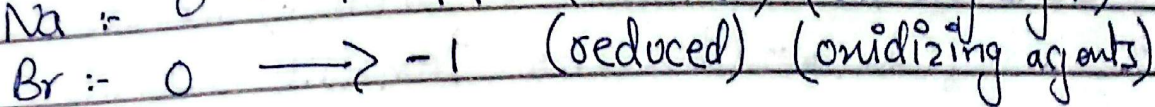
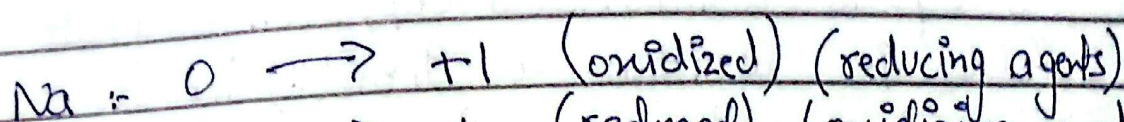
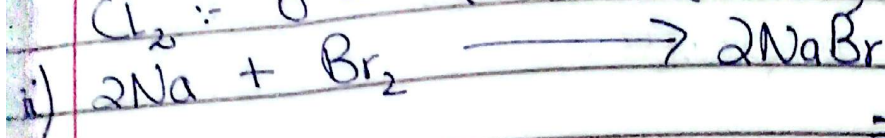
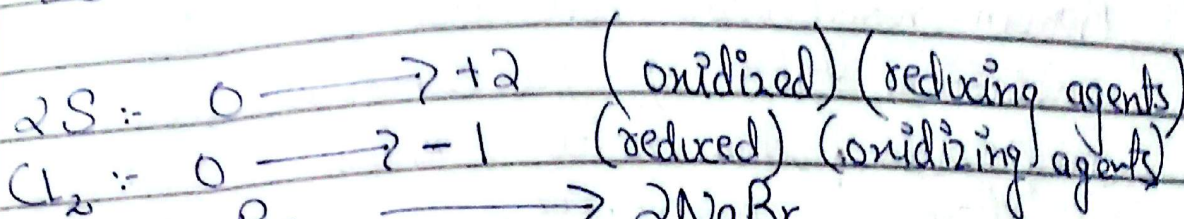
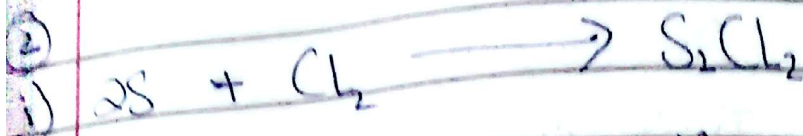
An oxidizing agent is the substance that causes another substance to oxidize by taking electrons from it. It is also called electron acceptor because it accepts electrons during reactions. Oxidizing agents are reduced in the process (they lose electrons).

### \* Reducing agents :-

A reducing agent is the substance that causes another substance to be reduced by donating electrons to it. It is also called electron donor because it donates electrons during reactions. Reducing agents are oxidized in the process (they gain electrons).

(P.T.O)







Q.3 (iii) (or) —

Ans.3 (iii) (or) —

①

Roman numerals are used to indicate the oxidation states of the elements in compounds, when metal exhibits variable oxidation states in compounds.

FOR EXAMPLE:-

⇒  $\text{CuSO}_4$  is written as copper(II) sulphate

⇒  $\text{FeCl}_2$  is written as iron(II) chloride.

⇒  $\text{FeCl}_3$  is written as iron(III) chloride.

②

① N in  $\text{NO}_2$

$$x + 2(-2) = 0$$

$$x - 4 = 0$$

$$x = 0 + 4$$

$$x = +4$$

②

S in  $\text{H}_2\text{SO}_4$

$$1(2) + x + 4(-2) = 0$$

$$2 + x - 8 = 0$$

$$x - 8 + 2 = 0$$



$$x + 6 = 0$$

$$x = +6 + 0$$

$$x = +6$$

iii) Cr in  $\text{Cr}_2\text{O}_3$

$$x(2) + 3(-2) = 0$$

$$2x - 6 = 0$$

$$2x = 0 + 6$$

$$2x = 6$$

$$x = \frac{6}{2}$$

$$x = +3$$

iv) Mn in  $\text{MnO}_4^-$

$$x + 4(-2) = -1$$

$$x - 8 = -1$$

$$x = -1 + 8$$

$$x = +7$$

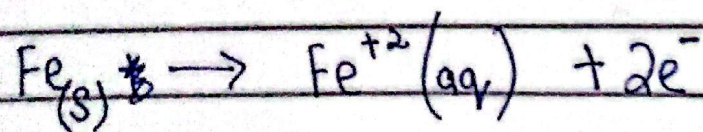


Q.3(iv)

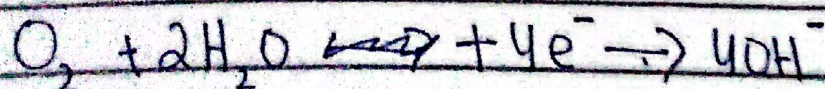
A:3(iv)

Corrosion is a natural electrochemical process in which a metal reacts with oxygen and moisture in the atmosphere. It is the electrochemical because it involves simultaneous oxidation (at the anodic region) and reduction (at the cathodic region), with electrons flowing between two regions through the metal.

Anodic half-reaction (less moisture region - oxidation):



Cathodic half-reaction (more moisture region - reduction):





$\text{Fe}^{2+}$  ions flow to cathodic region and reacts with  $\text{O}_2$  to form rust ( $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ )  
of preventing corrosion:-

① Coating with paint:-

Creates a physical barrier between the metal and the environment (oxygen and moisture). It is very cheap and easily applied.

② Alloying:-

Adding chromium to iron forms stainless steel, which is highly resistant to corrosion (chromium forms a self healing oxide layer)

③ Galvanizing:-

Coating iron with zinc, zinc acts both as a physical barrier and ~~as~~ as a sacrificial anode.

④ Electroplating:-

Using an electrolytic process to deposit a corrosion-resistant metal (e.g tin etc) on the iron surface.



## Section - B

Q: 2(i) (or)

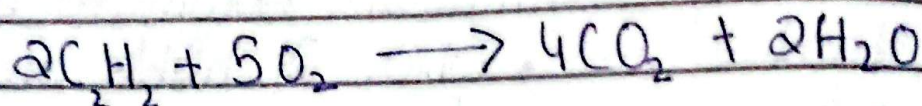
A: 2(i) (or)

Oxidation is the loss of hydrogen and reduction is the gain of hydrogen.

For Example:-

Acetylene is commercially used for cutting and welding metal. When it burns, it produces a very hot flame known as oxy-acetylene.

Equation:-



Q: 2(ii)

A: 2(ii)

Rules for assigning oxidation states:-

- 1) The oxidation numbers of any uncombined or free elements is always zero.
- 2) In simple ion, the oxidation number is same on their charge.
- 3) In complex ion, the sum of the oxidation number of atoms is equal to



their charges on their ions.

4) The oxidation number of each of the atoms in the molecule or compound is separately counted and their algebraic sum is zero.

① N in  $\text{HNO}_3$

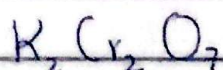
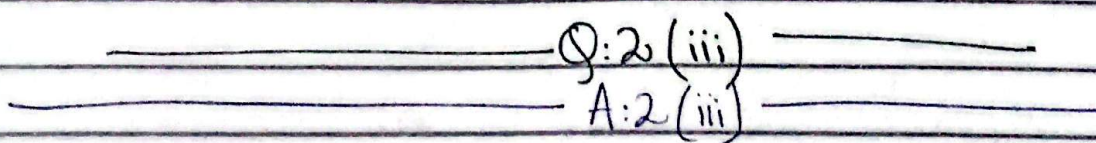
$$(1) + x + 3(-2) = 0$$

$$1 + x - 6 = 0$$

$$x - 6 + 1 = 0$$

$$x - 5 = 0$$

$$x = +5$$



$$1(2) + x(2) + 7(-2) = 0$$

$$2 + 2x - 14 = 0$$

$$2x - 14 + 2 = 0$$

$$2x - 12 = 0$$

$$2x = 12$$

$$x = \frac{12}{2}$$

$$\Rightarrow x = 6$$



Q: 2(iv)

A: 2(iv)

(i) N in  $\text{NO}_2$

$$x + 2(-2) = 0$$
$$x - 4 = 0$$
$$x = +4$$

(ii)  $\text{N}_2\text{O}$  ( $n=?$ )

$$2(x) + 1(-2) = 0$$
$$2x - 2 = 0$$
$$2x = 0 + 2$$
$$x = +1$$
$$x = +1$$

(iii) N in  $\text{HNO}_3$

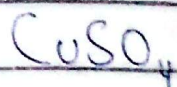
$$1 + x + 3(-2) = 0$$
$$1 + x - 6 = 0$$
$$x - 6 + 1 = 0$$
$$x - 5 = 0$$
$$x = +5 + 0$$
$$x = +5$$



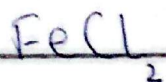
Q: 2(v) \_\_\_\_\_

A: 2(v) \_\_\_\_\_

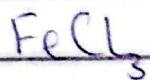
a) Copper (II) sulphate :- 



Iron (II) Chloride.

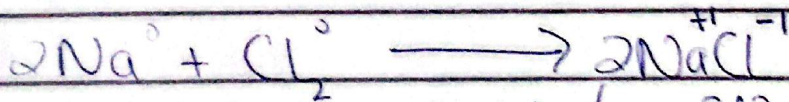


b) Iron (III) chloride



Q: 2(vi)(or) \_\_\_\_\_

A: 2(vi)(or) \_\_\_\_\_



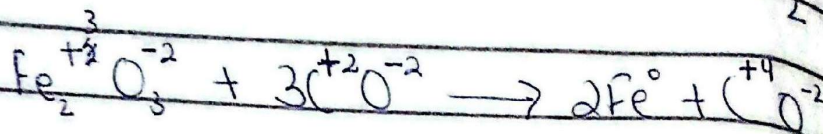
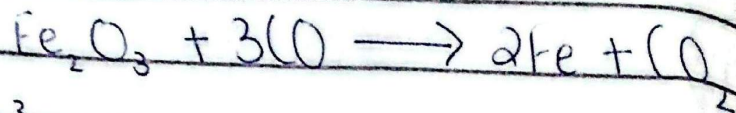
Na : 0  $\longrightarrow$  +1 (oxidized) (Reducing agent)

Cl<sub>2</sub> : 0  $\longrightarrow$  -1 (reduced) (oxidizing agent)

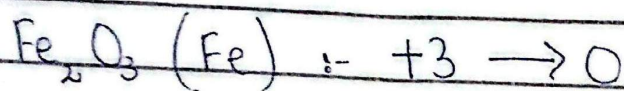


Q: 2 (vii)

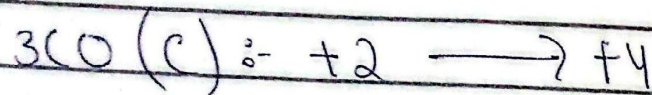
A: 2 (vii)



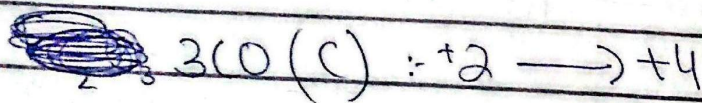
a) element ~~oxidized~~ <sup>reduced</sup> :-



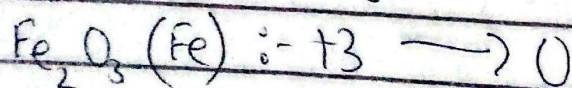
b) element oxidized :-



c) element ~~reducing~~ <sup>reducing</sup> :-



d) element oxidizing :-



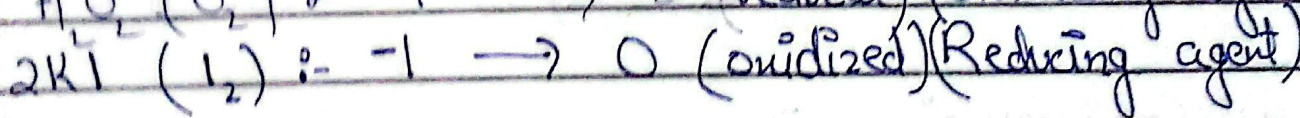
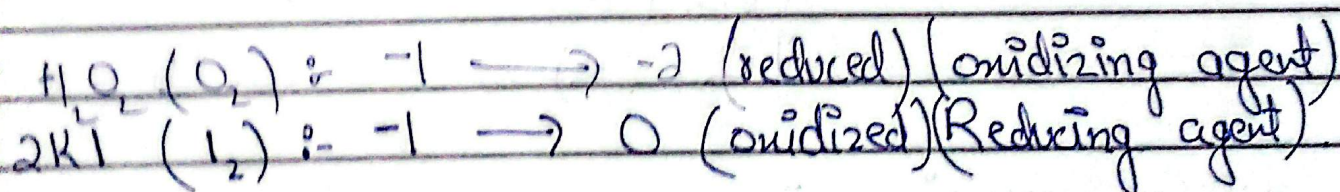
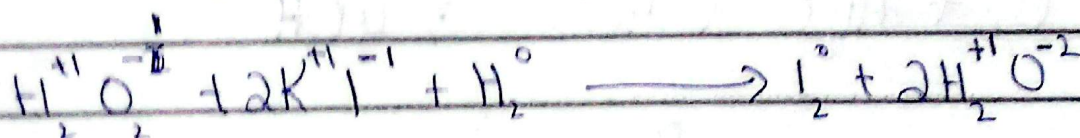
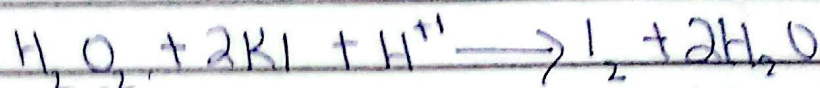


Q: 2 (viii)

A: 2 (viii)

When  $H_2O_2$  is added to potassium iodide (KI) solution, then the solution turns into reddish brown, indicating the formation of iodine ( $I_2$ ).

Equation:



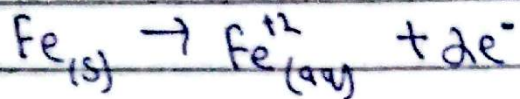
Q: 2 (ix)

A: 2 (ix)

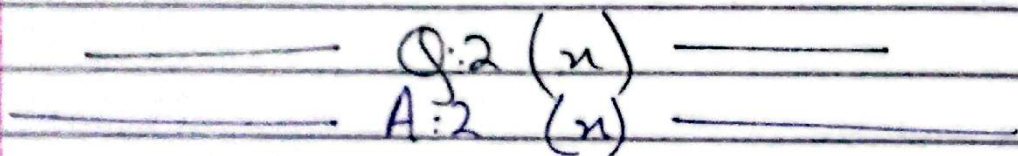
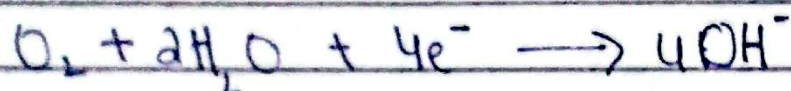
Corrosion is a natural electrochemical process in which a metal reacts with oxygen and <sup>moisture</sup> water to form rust. It is the electrochemical process because it involves simultaneous oxidation (at the anodic region) and reduction (at the cathodic region), with electrons flowing b/w two regions through the metal.



① Anodic - half reaction (less moisture) = oxidation - region):



Cathodic - half reaction (more moisture region - reduction).



① Galvanizing :-

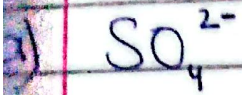
Coating iron <sup>with</sup> zinc, zinc acts both as a physical barrier and as a sacrificial anode.

② Electroplating :-

Using an electroplating electrolytic process to deposit a corrosion-resistant metal (e.g. tin, corrosion) on the ~~sur~~ iron surface.



Q: 2 (21)  
A: 2 (21)

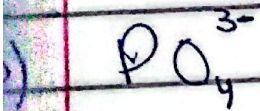


$$x + 4(-2) = -2$$

$$x - 8 = -2$$

$$x = -2 + 8$$

$$x = +6$$

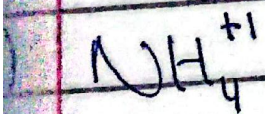


$$x + 4(-2) = -3$$

$$x - 8 = -3$$

$$x = +8 - 3$$

$$x = +5$$



$$x + 4(1) = +1$$

$$x + 4 = +1$$

$$x = +1 - 4$$

$$x = -3$$